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# Novel photocatalyst for dye degradation: $Cu_2O/Ag_2MoO_4$ nanocomposite on cellulose fibers from recycled cigarette butts

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ABSTRACT

Environmental and health concerns stemming from cigarette butt waste and water contamination by reactive dyes underscore the need for sustainable solutions. This study presented an innovative approach that repurposes discarded cigarette butts as a source of cellulose fibers (CFs). These CFs served as an eco-friendly substrate for the deposition of $Cu_2O/Ag_2MoO_4$ nanocomposite catalysts ($Cu_2O/Ag_2MoO_4$/CF), offering a sustainable and efficient method for the removal of reactive dyes, with specific focus on reactive blue 19 (RB19). The prepared heterostructure photocatalyst was thoroughly characterized using various analytical techniques, including XRD, UV–vis DR spectroscopy, SEM, TEM, TGA, photocurrent response, EIS measurements, PL spectroscopy, and FTIR spectroscopy. $Cu_2O/Ag_2MoO_4$/CF photocatalysts exhibited exceptional photodegradation performance toward RB19, surpassing monomeric and binary materials 3.6–7.6 times. Enhanced performance resulted from improved visible-light absorbance and efficient $e^-/h^+$ pair separation. Mechanistic insights gained from radical scavenger experiments identify hydroxyl radicals ($^{\cdot}OH$) and holes ($h^+$) as the primary active species responsible for photocatalysis. Reusability tests demonstrated the robustness and potential for multiple applications, maintaining a photocatalytic efficiency exceeding 86 % after five cycles, with negligible metal ion leaching. With superior visible-light photocatalytic performance, durability, and versatility, $Cu_2O/Ag_2MoO_4$/CF can hold significant potential for addressing challenges in wastewater treatment and environmental remediation, contributing to a cleaner and healthier environment.

## 1. Introduction

In recent years, escalating global environmental concerns have precipitated an intensified quest for sustainable remedies to combat the pressing issue of environmental pollution. Among the myriad of environmental contaminants, textile dyes have become a significant threat due to their persistence, toxicity, and harmful effects on ecosystems and human health [1]. Reactive dyes, a subgroup of vibrant organic compounds extensively used in textile coloring, form enduring covalent bonds with textile fibers, ensuring remarkable colorfastness. One such dye, Reactive Blue 19 (RB19; $C_{22}H_{16}N_2Na_2O_{11}S_3$), commonly known as Remazol Brilliant Blue R, is extensively used in coloring and printing cellulosic fibers [2]. Nevertheless, when wastewater containing RB19 contaminates natural ecosystems or is used for irrigation, it poses risks to aquatic life, plant health, and human well-being. Contact with RB19 has been linked to various health conditions, including skin allergies,

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contact dermatitis, occupational asthma, allergic conjunctivitis, and allergic rhinitis [3]. Consequently, there is a growing interest in exploring innovative and eco-friendly materials and technologies to efficiently remove these pollutants.

Photocatalytic methods have gained prominence in recent years as sustainable and efficient approaches for degrading organic pollutants, particularly textile dyes, in wastewater treatment. These methods are valued for their eco-friendliness, low energy consumption, high efficiency, and minimal generation of harmful byproducts [4]. Among the diverse semiconductors explored for their potential as photocatalytic materials, $Ag_2MoO_4$ has emerged as a noteworthy candidate due to its outstanding electrical conductivity, antimicrobial properties, luminescent characteristics, and photocatalytic activity. However, $Ag_2MoO_4$ primarily responds to ultraviolet light due to its wide bandgap (>3.0 eV), limiting its use in visible light photocatalysis [5]. To overcome this limitation, researchers have adopted strategies such as noble metal loading [6] and the fabrication of heterojunction structures [7,8] to broaden the light absorption range of semiconductors. Heterojunction structures, in particular, have proven highly effective, leading to significant advancements in photocatalytic performance. $Ag_2MoO_4$'s unique characteristic of having notably low conduction and valence band positions compared to other semiconductors has enabled the construction of heterojunction photocatalysts by combining it with various semiconductors [9]. This synergistic approach effectively separates photoinduced electron-hole pairs, extending their lifetime, and enhances the photocatalytic activity and stability of the resulting materials. Consequently, numerous heterojunction photocatalysts combining $Ag_2MoO_4$ with different semiconductors, such as $ZnIn_2S_4$/$Ag_2MoO_4$ [10], $Ag_2CrO_4$/$Ag_2MoO_4$ [11], MgS/$Ag_2MoO_4$ [12], $FeS_2$/$Ag_2MoO_4$ [13], CoS/$Ag_2MoO_4$ [14], $Ag_2MoO_4$/$Bi_4Ti_3O_{12}$ [9], Ag/$Ag_3PO_4$/$Ag_2MoO_4$ [15], $Fe_3O_4$-$C_3N_4$-$Ag_2MoO_4$ [16], have been successfully developed for the treatment of organic waste.

Copper(I) oxide ($Cu_2O$) is a p-type semiconductor with a narrow bandgap (1.8–2.5 eV) and a broad response range to solar spectrum [17]. $Cu_2O$ is considered a promising visible-light photocatalyst for constructing heterojunction photocatalysts aimed at achieving enhanced photocatalytic efficiency. It offers several advantages, including non-toxicity, environmental friendliness, low production costs, abundant resources, and high activity [18]. For example, Pahi *et al.* have integrated $Cu_2O$ with $Ag_3PO_4$ to create *p-n* heterojunctions for photocatalytic applications. The resulting $Ag_3PO_4$/$Cu_2O$ p-n heterojunction photocatalyst exhibited remarkable performance, with a hydrogen ($H_2$) generation rate of 737.4 μmol $g^{-1}h^{-1}$ and an endosulfan degradation efficiency of 91 %, surpassing pure $Ag_3PO_4$ by 12 and 9 times, respectively [19]. The p-n heterojunction photocatalyst demonstrated improved electron-hole migration efficiency and higher current density, synergistically enhancing catalytic activity through the interfacial space charge junction. Similarly, Gao *et al.* designed S-scheme heterojunction photocatalysts based on $Cu_2O$/BiOBr and studied their efficiency in photocatalytic degradation under saline solutions and real water conditions containing various organic dye contaminants [20]. This configuration enhanced the separation of photogenerated carriers and generated $^{\bullet}OH$ radicals through the interaction of Cu(I) and $H_2O_2$, resulting in efficient photocatalytic degradation.

In this context, the aim of this study is to enhance the photocatalytic efficiency for RB19 degradation by incorporating $Cu_2O$/$Ag_2MoO_4$ nanoheterojunctions. To further improve catalyst longevity and recyclability, a strategic immobilization onto cellulose fibers (CFs) derived from discarded cigarette butts (CBs) has been employed. This approach serves a dual purpose: preventing catalyst agglomeration to ensure sustained catalytic activity and enabling easy recovery, reducing material loss. Cellulose-supported metal nanoparticles have found extensive applications in green and sustainable chemical transformations, allowing for multiple recycling rounds without compromising their reactivity [21]. The choice of CBs waste as the source of CFs is based on the predominant presence of cellulose acetate, a primary constituent of CBs. This cellulose acetate can be readily converted into cellulose through alkaline treatment. Notably, approximately 6 trillion cigarettes are consumed worldwide annually, with approximately half of the CBs not disposed of properly [22]. These discarded butts, composed mainly of non-biodegradable plasticized cellulose acetate, toxic nicotine, and traces of heavy metals, present a significant environmental challenge. Repurposing them for environmental remediation aligns seamlessly with the principles of the circular economy and waste valorization, making a meaningful contribution to pollution control and resource conservation.

## 2. Materials and methods

### 2.1. Materials and chemicals

Silver nitrate ($AgNO_3$, 99.5 %), sodium molybdate dihydrate ($Na_2MoO_4{\cdot}2H_2O$, 99 %), ascorbic acid ($C_6H_8O_6$, 99.7 %), reactive blue 19 ($C_{22}H_{16}N_2Na_2O_{11}S_3$), potassium dichromate ($K_2Cr_2O_7$, ≥ 99 %), 4-p-benzoquinone (BQ, 99 %), isopropyl alcohol ($(CH_3)_2CHOH$, > 98 %), and ethylenediaminetetraacetic acid (EDTA, 99 %) were purchased from Merck. Copper sulfate pentahydrate ($CuSO_4{\cdot}5H_2O$, 99 %), ethylene glycol ($C_2H_6O_2$, 99 %), sodium hydroxide (NaOH, ≥ 98.0 %), hydrochloric acid (HCl, 37 %), and ethanol ($CH_3CH_2OH$, ≥ 99.8 %) were obtained from Sinopharm Chemical Reagent Company (Shanghai, China).

### 2.2. Characterisation techniques

The crystal structure was analyzed using X-ray diffraction (XRD) with a Rigaku Ultima IV diffractometer (Japan). Morphological characteristics were observed under a field emission scanning electron microscope (FE-SEM) S-4800 (Hitachi, Japan) and a transmission electron microscope (TEM) JEM 2100 (JEOL, Japan). Spectral data, including Photoluminescence (PL), Fourier-transform infrared spectroscopy (FTIR), and UV–visible diffuse reflectance spectroscopy (UV–vis DRS), were collected using various instruments: a Cary 4000 Eclipse Fluorescence Spectrometer (Agilent, USA), Bruker Tensor 27 spectrophotometer (Germany), and Shimadzu UV-2401 PC UV–Visible spectrophotometer (Japan). Thermogravimetric analysis (TGA) was performed on a LABSYS Evo TG-DTA instrument (USA) with a heating rate of 5 °C $min^{-1}$, covering the temperature range from 50 to 900 °C. Photocurrent curves and electrochemical impedance spectroscopy (EIS) Nyquist plots were acquired using a CHI 650E electrochemical workstation in conjunction with a three-electrode cell immersed in a $Na_2SO4$ (0.1 M) electrolyte solution. The concentration of RB19 was determined using a Cary 60 UV–Vis spectrophotometer (Agilent, USA). The extent of contaminant mineralization was evaluated by analyzing the total organic carbon (TOC) data utilizing an Analytikjena Multi N/S 2100S TOC analyzer (Germany). Metal ion concentrations in the solution were determined using inductively coupled plasma mass spectrometry (ICP-MS) with an Agilent 7700 instrument (USA).

### 2.3. Synthesis of photocatalysts

#### 2.3.1. Preparation of CFs

The CFs were prepared from CBs following a procedure outlined in our previous study [23]. In brief, the collected CBs were stripped of their outer paper wrapping, cut into approximately 1 mm-sized segments, and thoroughly washed with hot water. The CBs were then soaked in 50 % ethanol for 2 h to remove organic impurities and subsequently dried. Afterward, the dried CBs were immersed in a 5 % NaOH solution and heated to 60 °C for 4 h with continuous stirring. After treatment, the samples were filtered and rinsed with distilled water until reaching neutral pH. Finally, the resulting products were dried overnight at 60 °C.

### 2.3.2. Preparation of $Ag_2MoO_4$/CF

Initially, 0.1 g of CFs were dispersed in 30 mL of distilled water using ultrasonication for 10 min. A solution (Solution A) was prepared by adding 0.5 g of $AgNO_3$ to the CFs suspension and stirring for 1 h. Concurrently, another solution (Solution B) was created by dissolving 0.36 g of $Na_2MoO_4$ in 30 mL of distilled water. The $Ag_2MoO_4$/CF nanocomposite was formed by slowly adding Solution B to Solution A while continuously stirring for 4 h. After synthesis, the product was separated via centrifugation, washed with distilled water and ethanol to remove impurities, and then dried at 60 ℃ overnight.

### 2.3.3. Preparation of $Cu_2O/Ag_2MoO_4$/Cellulose fibers

To fabricate the $Cu_2O/Ag_2MoO_4$/CF nanocomposite, the as-synthesized $Ag_2MoO_4$/CF product was utilized as the starting material. First, it was immersed in a 50 mL solution of 0.1 M $CuSO_4$ and stirred at 60 °C for 1 h. Subsequently, a slowly added 50 mL mixture of (0.5 M NaOH + 0.25 M ascorbic acid) solution was introduced, and the mixture was further stirred for 4 h. The resulting product was separated by centrifugation, followed by thorough washing with distilled water and ethanol to remove any residual impurities. The nanocomposite was then dried at 60 °C overnight. The process for fabricating $Cu_2O/Ag_2MoO_4$/CF is visually depicted in Fig. 1. Additionally, to assess catalytic activity, comparative samples of $Cu_2O$, $Ag_2MoO_4$, and $Cu_2O/Ag_2MoO_4$ were also synthesized. Details of their synthesis processes are provided in the Supplementary Materials for reference.

## 2.4. Catalytic experimental procedures

The photocatalytic properties of the $Cu_2O/Ag_2MoO_4$/CF nanocomposite was evaluated for the degradation of RB19 dye. A quartz reactor was employed for the photocatalytic experiments, where the prepared composite (100 mg) was dispersed into an RB19 solution (100 mL, 25 mg/L). The reaction was performed at room temperature. Before irradiation, the suspension was stirred in the dark for 60 min to achieve adsorption–desorption equilibrium between the catalyst and the dye. Visible light, generated by a 250 W Xenon lamp equipped with a UV cutoff filter ($\lambda > 420$ nm), was then applied to the reaction mixture. Sampling at predefined time intervals (15 min) allowed for the monitoring of the degradation progress. The concentration of RB19 in the collected samples was determined using a Cary 60 UV–Vis spectrophotometer at 625 nm, enabling the assessment of degradation kinetics. Control experiments using $Cu_2O$, $Ag_2MoO_4$, $Cu_2O/Ag_2MoO_4$, and $Ag_2MoO_4$/CF followed the same procedure, providing comparative data for evaluating the photocatalytic activity of the $Cu_2O/Ag_2MoO_4$/CF nanocomposite. Key factors examined encompassed solution pH, which ranged from 3 to 9, catalyst dosage varying from 0.5 to 3 g/L, and RB19 concentration in the range of 15 to 45 mg/L. Data collected from these experiments were analyzed to ascertain the efficiency of the nanocomposite in degrading RB19 under visible light irradiation.

The degradation efficiency (DE%) and mineralization efficiency (ME, %) of RB19 were calculated according the Eq. (1) and (2), respectively.

$$\mathrm{DE}\,(\%) = \frac{C_0 - C_t}{C_0}\mathrm{x}100\% \quad (1)$$

$$\mathrm{ME}\,(\%) = \frac{TOC_0 - TOC_t}{TOC_0}\mathrm{x}100\% \quad (2)$$

where $C_0$ represents the initial RB19 concentration, and $C_t$ is the RB19 concentration at time $t$; $TOC_0$ and $TOC_t$ are the initial and final TOC values, respectively.

The kinetic degradation rates were determined by fitting the concentration–time data to pseudo-first-order kinetics using the formula $\ln(C_t/C_0) = -kt$, where $k$ represents the apparent rate constant.

## 3. Results and disccusion

### 3.1. Characterization of photocatalysts

The XRD patterns of the examined materials, including pure CF, $Cu_2O$, $Ag_2MoO_4$, $Ag_2MoO_4$/CF, and $Cu_2O/Ag_2MoO_4$/CF, were analyzed to elucidate their crystallographic properties, as presented in Fig. 1a. The XRD pattern of the CF sample exhibits distinct peaks at 2θ values of

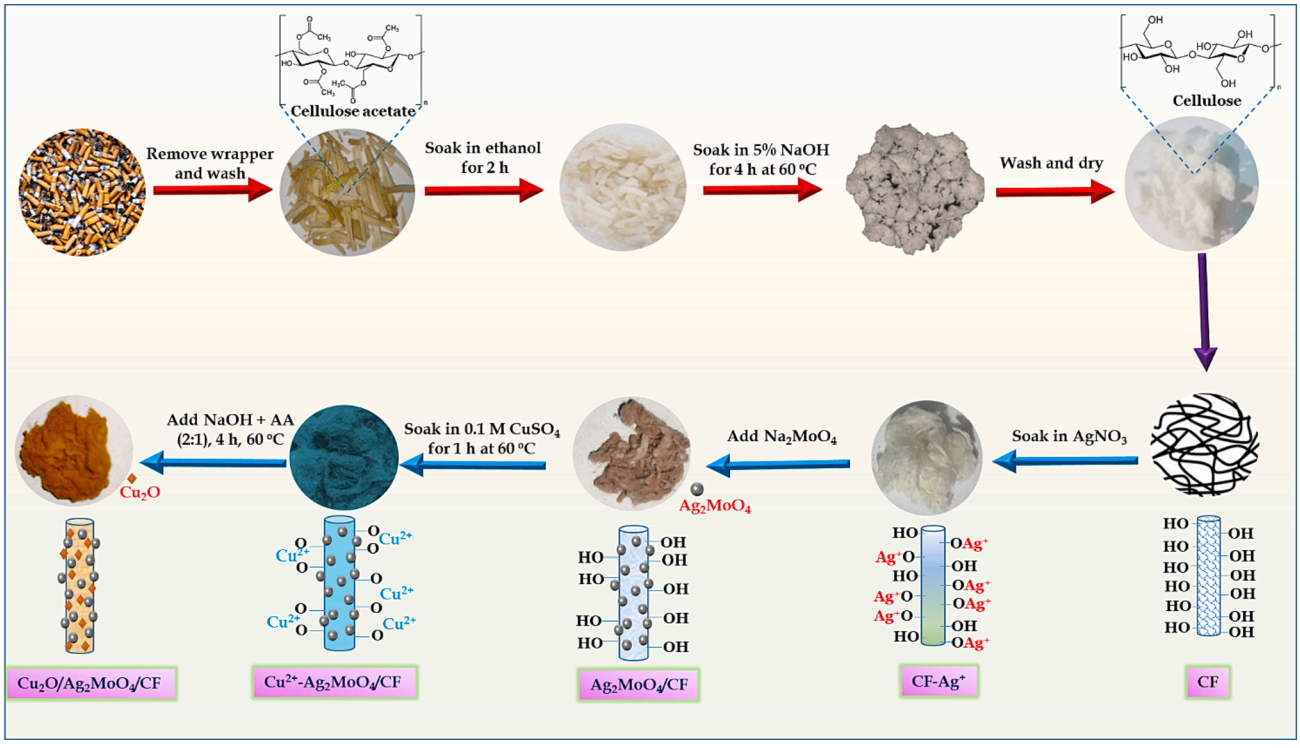



**Fig. 1.** Schema for synthesizing $Cu_2O/Ag_2MoO_4$/CF composite.

12.3°, 20.4°, 21.2°, and 29.6°, which are characteristic of cellulose II [24]. These diffraction peaks exhibit relatively low intensity, suggesting the amorphous nature of CF. Additionally, an intense peak at 25.9° is observed in the XRD pattern of CF, potentially attributed to carbon (002), likely resulting from cellulose degradation induced by NaOH treatment. For $Cu_2O$, the XRD pattern displays well-defined diffraction peaks at 2θ angles of approximately 29.5°, 36.43°, 36.4°, 42.3°, 61.3°, 73.5°, and 77.4°, corresponding to the (101), (111), (200), (220), (311), and (222) crystallographic planes, respectively. These peaks are consistent with the $Cu_2O$ phase (PDF Card No. 05–0667) [20]. The sharp and resolved $Cu_2O$ XRD peaks affirm the high purity of the synthesized $Cu_2O$ sample, with minimal presence of copper oxides or residual reactants. In the case of $Ag_2MoO_4$, the XRD pattern reveals characteristic peaks at 2θ angles of 16.5°, 27.0°, 31.8°, 33.3°, 38.6°, 42.3°, 47.8°, 50.8°, 55.8°, 63.1°, 65.6°, 66.6°, 69.9°, 72.3°, 76.5°, and 78.8°, corresponding to the (110), (220), (311), (222), (400), (331), (422), (511), (440), (620), (533), (622), (444), (711), (642), and (731) planes, respectively. These diffraction peaks align with the crystal structure of $Ag_2MoO_4$ (JCPDS No. 08–0473) [25]. In the ternary $Cu_2O$/$Ag_2MoO_4$/CF nanocomposite, the XRD pattern retains the characteristic diffraction peaks of CF, $Cu_2O$, and $Ag_2MoO_4$, confirming the successful formation of the composite material. The observed peaks at 2θ angles corresponding to $Cu_2O$ and $Ag_2MoO_4$ indicate that the individual phases are well-preserved within the composite, signifying excellent dispersion and interaction between the components.

The FTIR analysis results reveal the presence of distinct functional groups in the tested materials (Fig. 2b). In the CFs sample, the FTIR spectrum exhibits characteristic bands at wavenumbers of 3448, 1645,

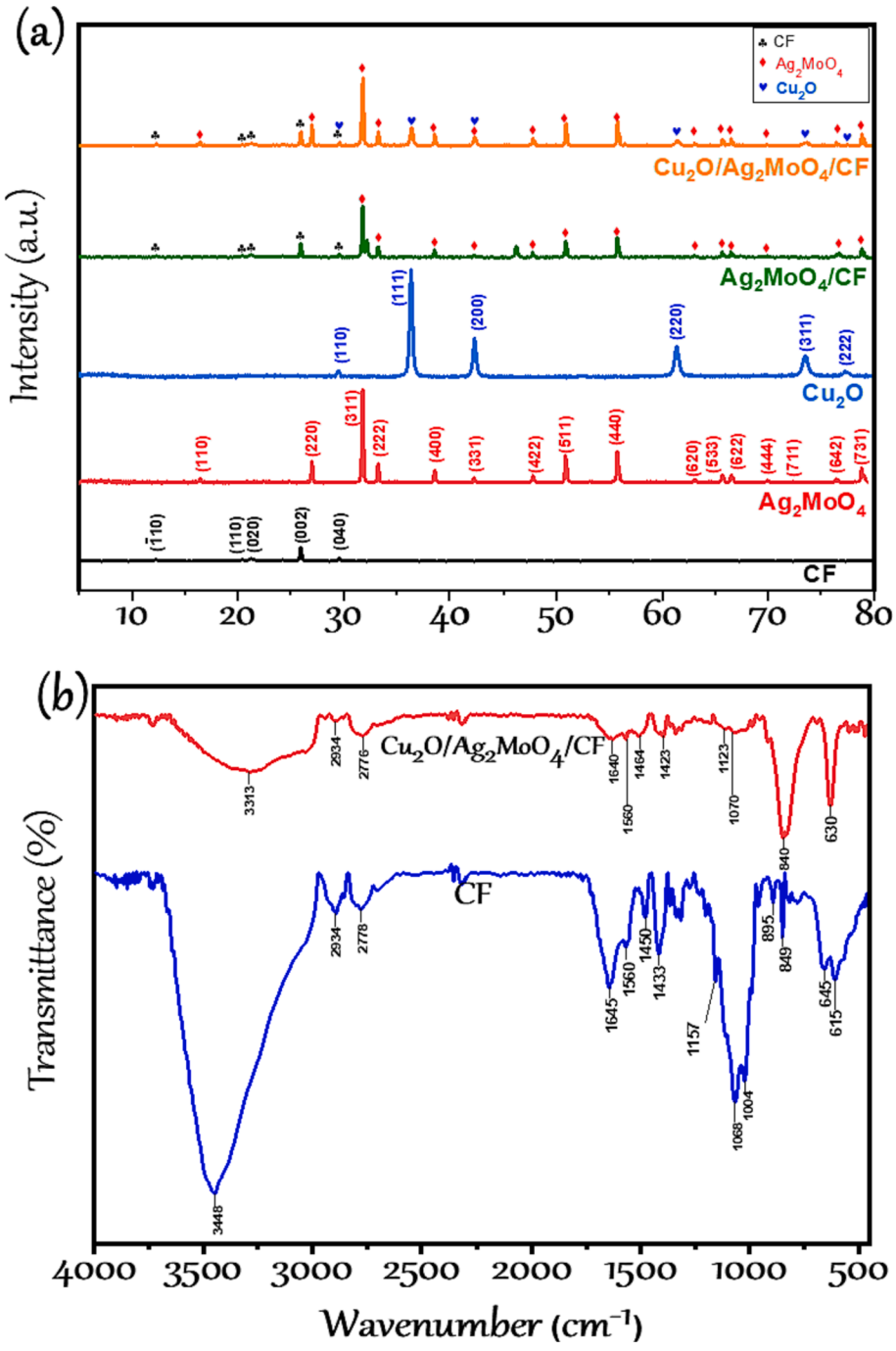



**Fig. 2.** XRD pattern (a) and FTIR spectra (b) of the synthesized samples.

1068, and 1004 $cm^{-1}$, corresponding to the hydroxyl (–OH), carbonyl (C = O), acetyl ester ($CH_3$–C = O), and ether (C–O–C) groups, respectively [23]. Additional peaks are observed at 2934 $cm^{-1}$ and 2778 $cm^{-1}$, representing the symmetric and asymmetric stretching vibrations of methyl groups (C–H). A peak at 1433 $cm^{-1}$ corresponds to C–H asymmetric deformation, while the peaks at 1157 $cm^{-1}$ can be assigned to C–O stretching. Notably, the absorption band at 898 $cm^{-1}$ is characteristic of the β-glycosidic linkage between glucose units [22]. Upon analyzing the FTIR spectrum of the $Cu_2O/Ag_2MoO_4$/CF composite, similarities with the CFs spectrum are evident, albeit with variations in the intensity and location of functional groups. Specifically, the –OH, –C = O, –$CH_3$–C = O, C–O–C, as well as –C–H and –$CH_2$ groups all experience a notable decrease in intensity in this composite spectrum. This reduction suggests interactions and chemical bonding between $Cu_2O$, $Ag_2MoO_4$, and cellulose within the nanocomposite structure. Distinctive peaks at 630 $cm^{-1}$ are assigned to the Cu–O group, while the band at 840 $cm^{-1}$ corresponds to the O–Mo–O group, which aligns with previous reports [20,26]. The presence of these peaks indicates that $Cu_2O$ and $Ag_2MoO_4$ have combined with cellulose, forming specific chemical bonds within the nanocomposite system.

The FE-SEM analysis of CFs, $Ag_2MoO_4$, $Cu_2O$, and $Cu_2O/Ag_2MoO_4$/CF samples provides valuable insights into their morphological characteristics (Fig. 3). In the FE-SEM micrograph of CF (Fig. 3a), the material appears as a network of fibers with an approximate diameter of 15 μm, interconnected with each other. The surface of CF exhibits a rugged structure with numerous ridges and folds, indicating its fibrous and porous nature. Fig. 3b displays FE-SEM images of the $Ag_2MoO_4$ sample, revealing that the $Ag_2MoO_4$ particles are observed as individual potato-like-shaped entities with smooth surfaces. The size of the $Ag_2MoO_4$ particles varies within the range of 1–3 μm. Similar morphological characteristics of $Ag_2MoO_4$ have also been reported in previous publications [27,28]. In Fig. 3c, SEM micrographs of $Cu_2O$ showcase octahedral-shaped particles clearly identified with sizes ranging from 75 to 150 nm. The even distribution of $Cu_2O$ particles indicates a consistent and controlled synthesis process. It's worth noting that octahedral $Cu_2O$ with exposed (1 1 1) facets is known to exhibit significantly higher photocatalytic activity compared to the cubic structure [29]. FE-SEM images of the $Cu_2O/Ag_2MoO_4$/CF nanocomposite in Fig. 3d-f demonstrate the successful deposition of $Cu_2O$ and $Ag_2MoO_4$ particles onto the CF substrate. The composite exhibits heterogeneous morphology with the presence of potato-like $Ag_2MoO_4$ particles and octahedral $Cu_2O$ particles adhering to the CF network. It is noteworthy that the size of $Ag_2MoO_4$ in the composite is much smaller (<200 nm) than when synthesized separately (>1 μm). This phenomenon may be attributed to the *in-situ* growth of $Ag_2MoO_4$ crystals on the CF support with more nucleation sites. This enhances the catalytic activity of the composite by increasing the surface area.

The TEM technique was further employed to scrutinize the morphology of catalyst nanoparticles anchored onto CFs. Fig. 4a offers a clear visual representation of nano $Cu_2O$ and $Ag_2MoO_4$ particles firmly adhering to the CFs substrate. Furthermore, Fig. 4b unveils the precise dimensions and configurations of these catalyst nanoparticles. The $Ag_2MoO_4$ crystals adopt a distinctive potato-like morphology, while $Cu_2O$ takes on an octahedral form, echoing the observations made through FE-SEM. The HR-TEM analysis unequivocally validates the presence of $Ag_2MoO_4$ and $Cu_2O$, characterized by distinctive d-spacing values, which were determined using ImageJ software. Specifically, a d-spacing of 0.27 nm corresponds to the (3 1 1) crystallographic plane of $Ag_2MoO_4$, while a d-spacing of 0.24 nm corresponds to the (1 1 1) crystallographic plane of $Cu_2O$. These findings are in concordance with prior researches [30,31].

The elemental composition and spatial distribution of the $Cu_2O/Ag_2MoO_4$/CF composite were comprehensively analyzed through EDX and element mapping techniques, as illustrated in Fig. 5. The EDX spectrum quantitatively characterizes the composite's elemental composition, identifying carbon (C), oxygen (O), copper (Cu), molybdenum (Mo), and silver (Ag) with respective weight percentages of 42.45 %, 40.93 %, 12.02 %, 1.45 %, and 3.15 %. Elemental mapping images vividly depict the dispersion of these elements across the composite material. These mappings visually confirm the integration of these elements within the $Cu_2O/Ag_2MoO_4$/CF sample, showcasing their even distribution and uniform dispersion throughout the composite's microstructure. These findings not only substantiate the successful synthesis of $Cu_2O/Ag_2MoO_4$/CF but also validate the presence of its constituent elements, consistent with SEM and HR-TEM observations.

The UV–Vis DRS spectra (Fig. 6a) were analyzed to differentiate the optical properties of the prepared materials and estimate the corresponding bandgap energies. As illustrated in the spectra, each material exhibits characteristic absorption patterns in the UV and visible regions. The UV–Vis DRS spectrum of CF primarily shows absorption in the UV region, with an absorption edge at 352 nm, which is typical for organic materials. In contrast, bare $Ag_2MoO_4$ exhibits significant light absorption in the UV and visible regions, with an absorption edge at 388 nm. However, when combined with CF ($Ag_2MoO_4$/CF composite), its light-absorbing capacity is reduced. $Cu_2O$ on its own exhibits notably high light absorption, particularly in the 400–500 nm range, visible as a sharp

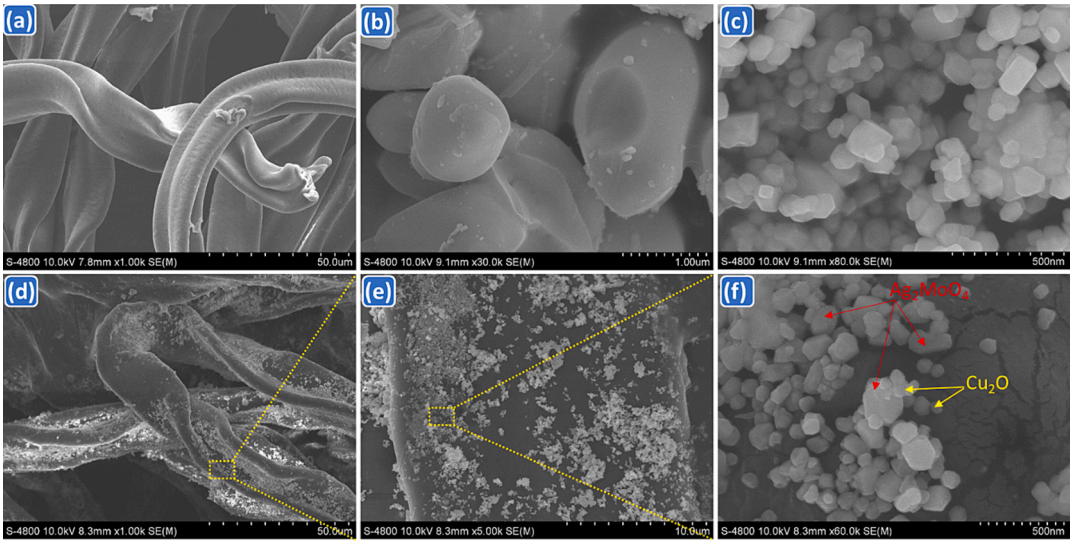



**Fig. 3.** FE-SEM images of (a) CFs, (b) $Ag_2MoO_4$, (c) $Cu_2O$, (d, e, f) $Cu_2O/Ag_2MoO_4$/CF.

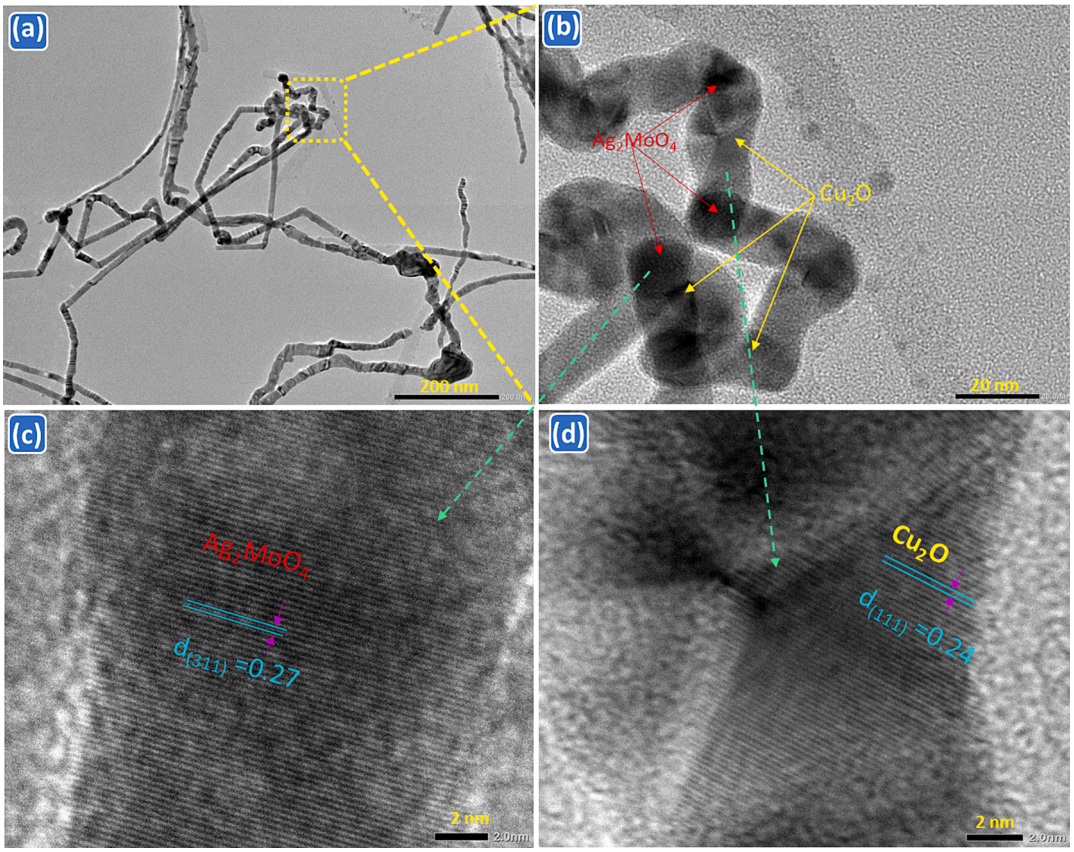



**Fig. 4.** TEM images (a, b) and HR-TEM images (c, d) for $Cu_2O/Ag_2MoO_4/CF$.

absorption peak. The ternary $Cu_2O/Ag_2MoO_4/CF$ composite demonstrates enhanced light absorption in both the ultraviolet and visible ranges compared to bare $Ag_2MoO_4$ and the binary composites $Ag_2MoO_4/CF$ and $Cu_2O/Ag_2MoO_4$. The light absorption behavior of the ternary composite and bare $Cu_2O$ is quite similar in the wavelength range below 520 nm. However, at longer wavelengths, the composite exhibits superior light absorption compared to $Cu_2O$ alone, attributed to the synergistic effect of $Ag_2MoO_4$.

The bandgap energy (Eg) of a semiconductor materials was determined by the Tauc method using the Eq (3) [14]:

$$(\alpha h\nu)^2 = A(h\nu - Eg) \tag{3}$$

where, α is the absorption coefficient, h is Planck's constant, ν is the frequency of light, A is the arbitrary constant, and Eg is the bandgap energy. By plotting $(\alpha h\nu)^2$ versus hν, the bandgap energy can be calculated from the intercept of the linear portion of the curve (Fig. 6b). The Eg values for CF, $Cu_2O$, $Ag_2MoO_4$, $Ag_2MoO_4/CF$, $Cu_2O/Ag_2MoO_4$, and $Cu_2O/Ag_2MoO_4/CF$ were determined to be 3.44, 2.17, 3.18, 3.39, 2.23, and 2.17 eV, respectively. The reduction in the composite's band gap can be explained by its extensive absorption of visible light, resulting in an increased concentration of charge carriers within its transition region [19]. The band edge potential of the pure component was further calculated Mulliken-electronegativity model according the Eq. (4) and (5).

$$E_{VB} = \chi - E_e + 0.5Eg \tag{4}$$

$$E_{CB} = E_{VB} - Eg \tag{5}$$

In this context, $E_{VB}$ represents the energy level of the valence band (VB), $E_{CB}$ represents the energy level of the conduction band (CB), and Eg denotes the band-gap energy. Additionally, χ signifies the absolute electronegativity value of the semiconducting material, which was 5.32 for $Cu_2O$ [32] and 5.92 for $Ag_2MoO_4$ [25]. The reference energy of the free electron (Ee) was set at 4.5 eV, relative to the hydrogen scale. The calculated energy band positions for $Cu_2O$ were −0.26 eV (CB) and + 1.91 eV (VB), while for $Ag_2MoO_4$, they were −0.17 eV (CB) and + 3.01 eV (VB). These results align with prior research findings [19,25,33].

Photoluminescence spectroscopy was employed to assess the electron-hole recombination behavior of the prepared materials. It is a well-established fact that the PL spectrum of semiconductor materials predominantly originates from the recombination of photo-induced charge carriers. Generally, a lower PL intensity signifies a reduction in the rate of charge carrier recombination [34]. As depicted in Fig. 6c, the PL spectrum of the $Ag_2MoO_4$ sample displayed three peaks at 485 nm, 530 nm, and 542 nm, corresponding to specific electronic transitions within the $MoO_4$ tetrahedra [35]. When combined with CFs, a noticeable decrease in the PL intensity of $Ag_2MoO_4$ was observed. This reduction in PL intensity can be attributed to several factors. First, the presence of CFs may act as electron transport pathways, facilitating the migration of charge carriers and reducing their recombination rate, ultimately leading to a decrease in PL emission. Second, CFs may effectively quench the excitons formed within $Ag_2MoO_4$, further contributing to the decrease in PL emission. Furthermore, with the addition of $Cu_2O$ to form the composite $Cu_2O/Ag_2MoO_4/CF$, there was a further reduction in PL intensity compared to $Ag_2MoO_4/CF$. This observation demonstrates that the level of electron-hole recombination is significantly constrained due to the heterojunction [36]. The decrease in the PL intensity suggests that the inclusion of $Cu_2O$ enhances the charge carrier separation within the composite, resulting in fewer excitons reaching the radiative recombination pathways, and consequently, a decrease in PL emission.

The photo-generated electron and hole transfer efficiency, as well as spatial separation, were further investigated through transient photocurrent and EIS analyses. Fig. 6d illustrates the photocurrent responses

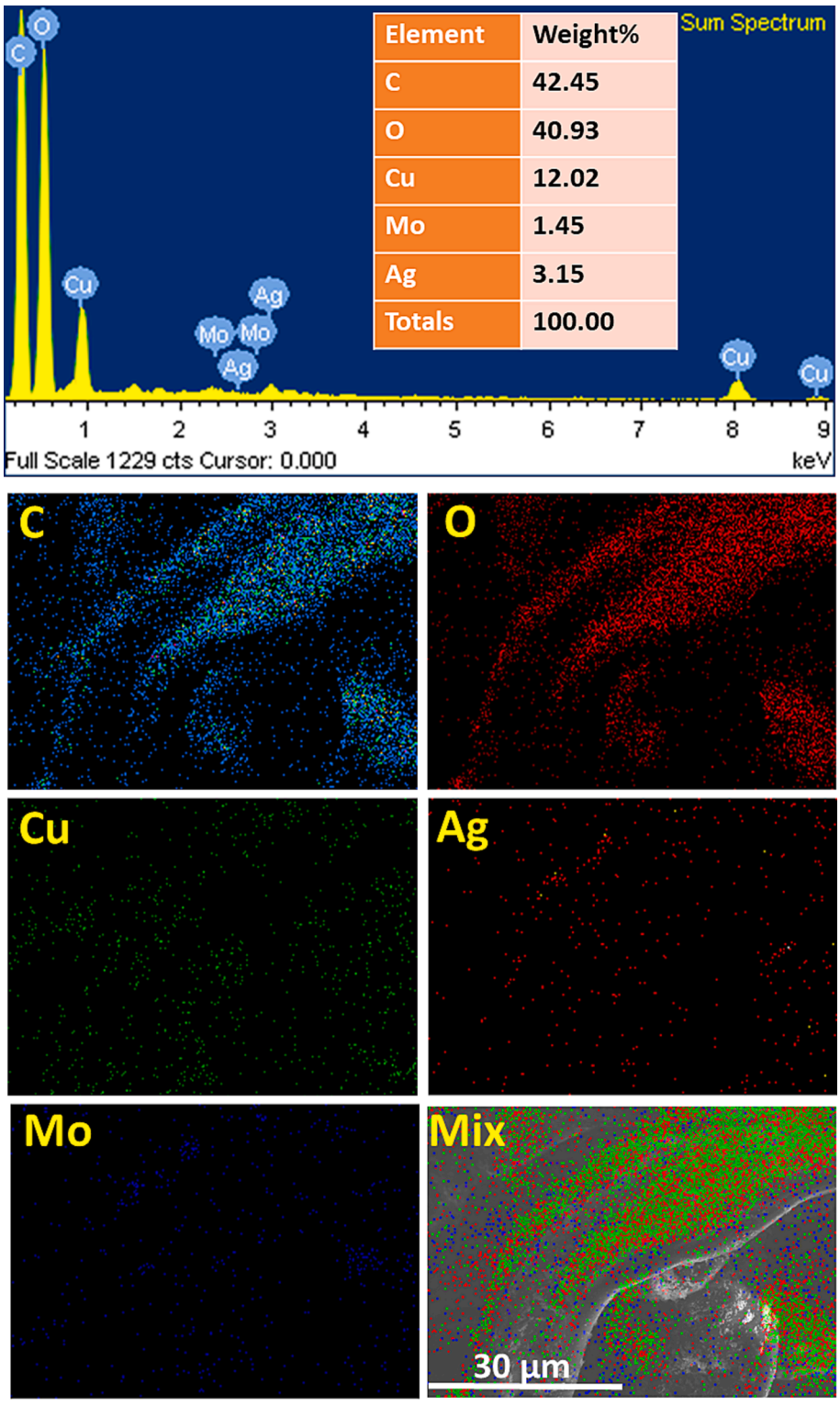



**Fig. 5.** EDX spectra, chemical composition and element mapping of $Cu_2O/Ag_2MoO_4$/CF sample.

of the prepared samples under visible light, providing insights into the efficiency of charge transfer and separation. The results demonstrate the excellent stability and consistency of the photocurrent generation, evident from the consistent on/off responses to light stimuli. The photovoltaic intensity of the $Cu_2O/Ag_2MoO_4$ composite outperformed that of individual $Ag_2MoO_4$ and $Cu_2O$. Notably, the $Cu_2O/Ag_2MoO_4$ composite exhibited a significantly higher photocurrent, measuring 3.33 and 7.69 times greater than that of $Cu_2O$ and $Ag_2MoO_4$, respectively. This enhancement suggests the superior capability of the $Cu_2O/Ag_2MoO_4$ composite in effectively separating photo-induced electron-hole pairs, predominantly attributed to the formation of heterojunctions between the two materials [37]. To further characterize the charge transfer properties, the EIS Nyquist plot of $Cu_2O$, $Ag_2MoO_4$, and $Cu_2O/Ag_2MoO_4$ composites is presented in Fig. 6e. Notably, the nano-composite material comprising $Cu_2O/Ag_2MoO_4$ exhibited the shortest semicircular radius on the EIS spectrum. The semicircle radius is often

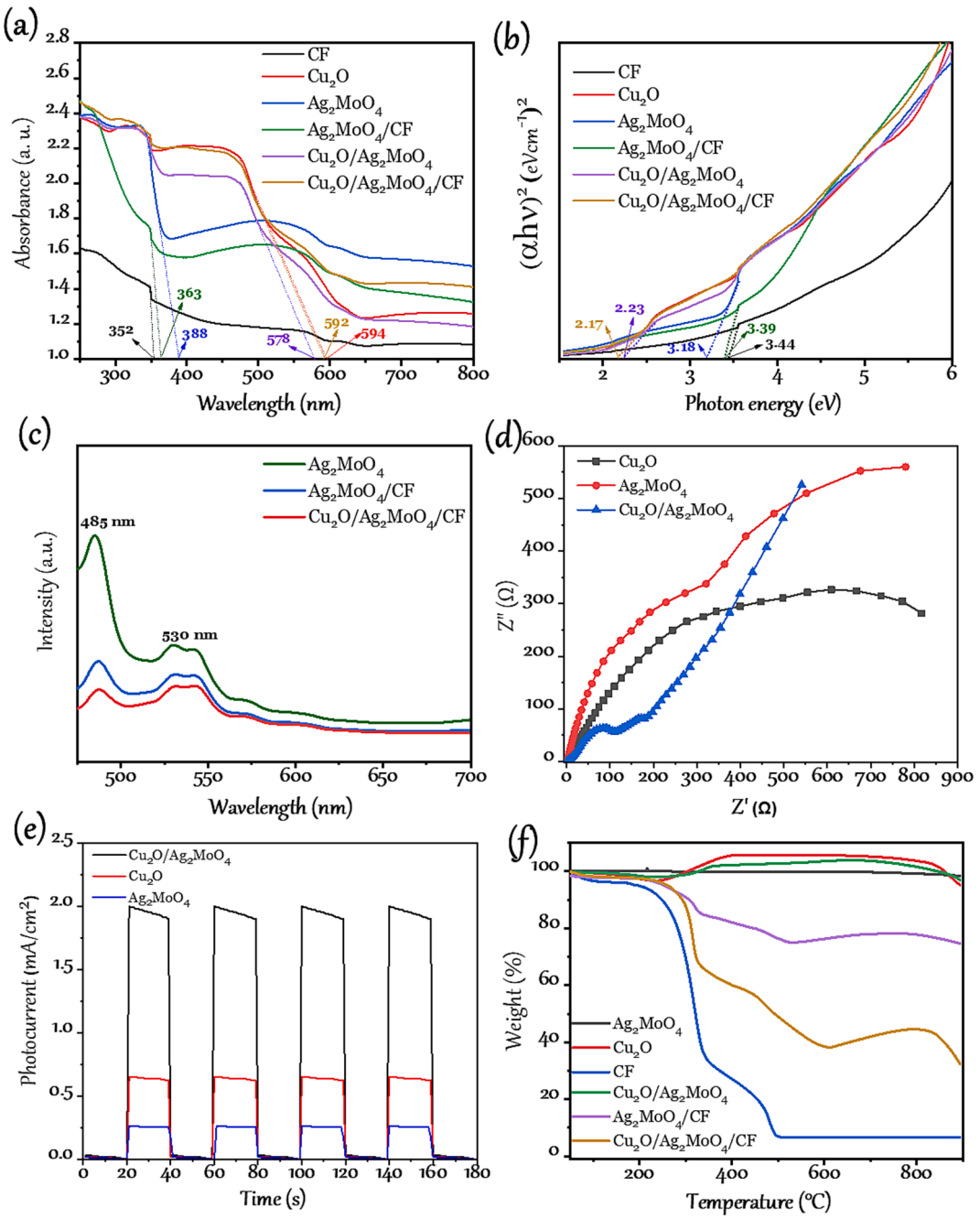



**Fig. 6.** UV–vis-DRS spectra (a), Tauc's plots (b), PL spectra excitated at wavelength of 400 nm (c), transient photocurrent response curves (d), EIS spectra (e), and TG curves (f) for as-synthesized samples.

associated with charge transfer resistance, where a smaller radius indicates more facile electron movement and efficient charge transfer [38]. Therefore, the EIS results strongly support the notion that the combination of $Cu_2O$ and $Ag_2MoO_4$ facilitates the effective separation and transfer of electron-hole pairs generated on the surface, consistent with the transient photocurrent response. While the transient photocurrent and EIS analyses were not extended to include the $Cu_2O$/$Ag_2MoO_4$/CF composite due to challenges in securing cellulose fiber adhesion, it is reasonable to anticipate that the $Cu_2O/Ag_2MoO_4$/CF composite could potentially exhibit enhanced charge transfer properties. The synergistic effects of cellulose fibers, serving as a substrate, and the $Cu_2O/Ag_2MoO_4$ heterojunction may contribute to improved electron movement and efficient charge separation, suggesting promising prospects for the overall photoelectrochemical behavior of the composite. These findings underscore the enhanced photoelectrochemical performance of the $Cu_2O/Ag_2MoO_4$ and $Cu_2O/Ag_2MoO_4$/CF composite, affirming its potential as a novel photocatalyst for dye degradation.

The stability of the synthesized materials was assessed through thermogravimetric analysis in the temperature range of 50–900 °C, as depicted in Fig. 6f. The CFs sample exhibit notable resilience at temperatures below 200 °C, experiencing only a marginal 6 % reduction in initial mass. Beyond 200 °C, CFs undergo substantial decomposition, with nearly 94 % mass loss observed by 500 °C, indicating their vulnerability to elevated temperatures. In contrast, $Ag_2MoO_4$ demonstrates remarkable thermal stability, evidenced by a mere 2 % mass loss across the entire temperature range from 50 °C to 900 °C. For $Cu_2O$, a modest mass reduction of approximately 3.5 % is observed up to 250 °C, likely associated with the release of adsorbed water within the sample.

Intriguingly, a distinct weight gain (about 5 % compared to the initial weight) is observed in the 250 – 850 °C range for $Cu_2O$. This weight increase is attributed to the presence of residual air between the fine powder particles, as the loosely structured synthesized materials can readily adsorb gases and air. Consequently, this leads to the oxidation of $Cu_2O \rightarrow CuO$ transformation. This phenomenon has been reported in studies by Alekseeva *et al.* [39]. The transformation also contributes to the observed weight gain in samples containing $Cu_2O$, such as $Cu_2O/Ag_2MoO_4$ and $Cu_2O/Ag_2MoO_4$/CF. Another noteworthy finding in all $Cu_2O$-containing samples in this study is the initiation of mass reduction at temperatures higher than 850 °C. The reason given is the reversible decomposition of CuO into $Cu_2O$ [40]. The mass loss in this temperature range for the $Cu_2O/Ag_2MoO_4$/CF sample is the highest, as it involves the additional reduction of CuO and $Cu_2O$ to Cu by carbon generated from the decomposition of CF. The thermal behavior of the $Cu_2O/Ag_2MoO_4$ sample closely resembles that of $Cu_2O$, with a slight weight increase occurring at higher temperatures (330 °C) due to the presence of thermally stable $Ag_2MoO_4$. The $Cu_2O/Ag_2MoO_4$ and $Cu_2O/Ag_2MoO_4$/CF composites exhibit similar thermal behaviors at temperatures below 285 °C, displaying almost negligible mass loss up to 250 °C (approximately 4 % mass loss). However, at temperatures exceeding 300 °C, $Cu_2O/Ag_2MoO_4$/CF experiences greater mass loss due to the thermal decomposition of CFs. Upon heating to 900 °C, $Cu_2O/Ag_2MoO_4$ and $Cu_2O/Ag_2MoO_4$/CF exhibit mass reductions of 25 % and 68 %, respectively.

### 3.2. Photocatalytic activity

The photocatalytic activity of various fabricated photocatalysts was assessed in the visible-light degradation of RB19. Fig. 7a illustrates the comparative degradation of RB19 by $Cu_2O$, $Ag_2MoO_4$, $Ag_2MoO_4$/CF, $Cu_2O/Ag_2MoO_4$, and $Cu_2O/Ag_2MoO_4$/CF. Evidently, photolysis of RB19 exhibited negligible degradation, with only approximately 3 % of RB19 removed over 150 min of visible light irradiation, indicating RB19's relative stability under visible light. After 150 min of visible light exposure, the RB19 degradation efficiency was 49.1 %, 60.3 %, 66.7 %, 74.3 %, and 98.9 % in the presence of $Ag_2MoO_4$, $Ag_2MoO_4$/CF, $Cu_2O$, $Cu_2O/Ag_2MoO_4$, and $Cu_2O/Ag_2MoO_4$/CF, respectively. These results highlight that the incorporation of $Cu_2O$ significantly enhances the catalytic activity of $Ag_2MoO_4$, primarily due to the reduction in the recombination rate of electron-hole pairs. Notably, the ternary $Cu_2O/Ag_2MoO_4$/CF composite exhibited superior RB19 degradation compared to $Cu_2O$, pristine $Ag_2MoO_4$, or $Cu_2O/Ag_2MoO_4$. The attachment of the $Cu_2O/Ag_2MoO_4$ nanocomposite onto CFs clearly proves to be effective due to the creation of an efficient solid–solid interface between $Cu_2O$ and $Ag_2MoO_4$ on the CFs' surface. This interface is where the separation of photo-induced electrons and holes occurs most effectively, preventing their recombination and providing favorable conditions for complex photochemical reactions during RB19 degradation. By being immobilized on the CFs, $Cu_2O$ and $Ag_2MoO_4$ nanoparticles can interact closely, allowing for efficient charge transfer and preventing the rapid

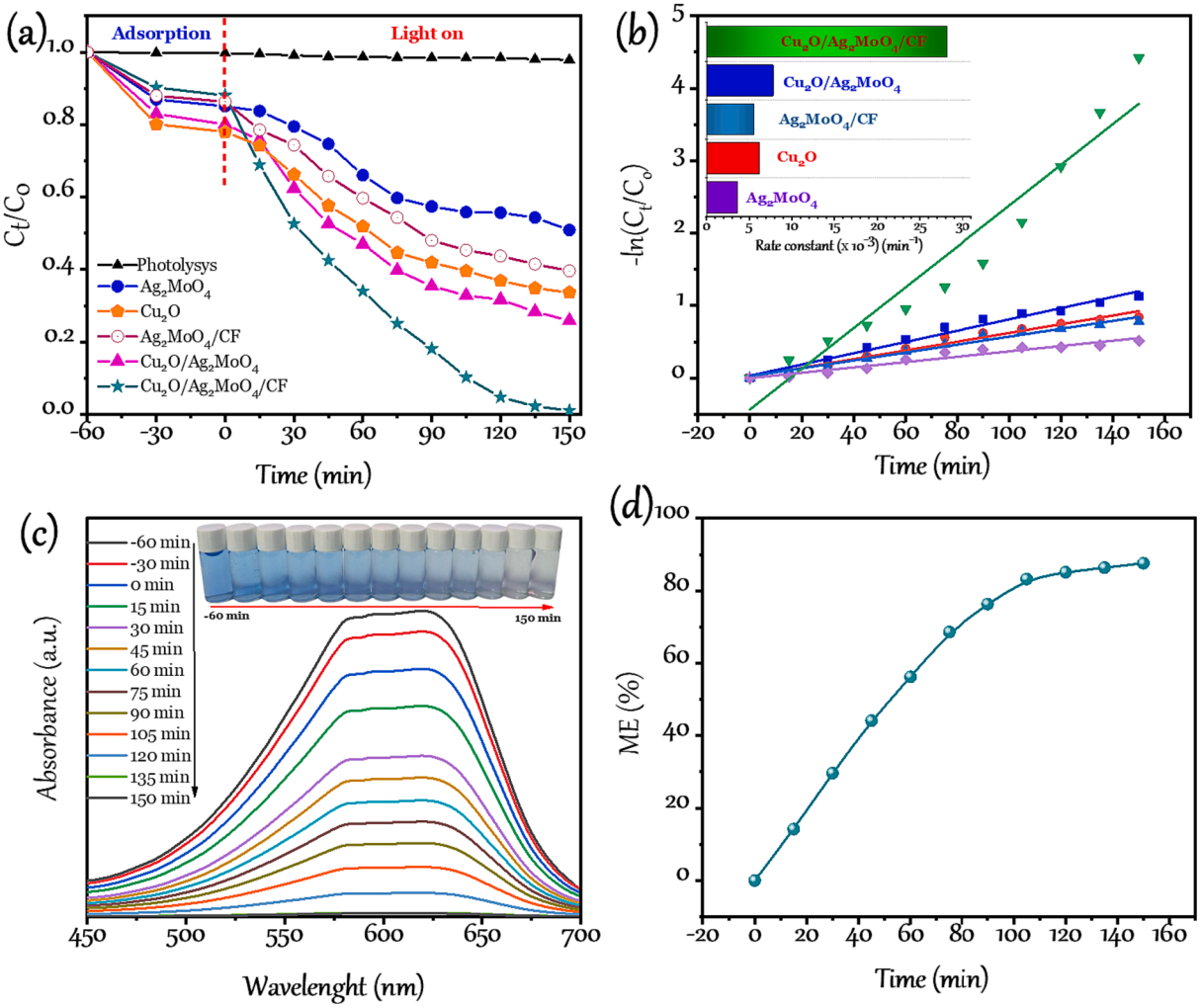



**Fig. 7.** Photocatalytic activity (a) and rate constants of photocatalysts (b), UV–Vis spectra of RB19 solution (c) and ME (d) during photodegradation by $Cu_2O/Ag_2MoO_4$/CF. Experimental condition: [RB19] = 25 mg/L, catalyst dosage of 1 g/L, 25 °C, pH 7.

recombination of electrons ($e^-$) and holes ($h^+$) [41]. This phenomenon provides highly favorable conditions for complex photochemical reactions during the degradation of RB19.

The degradation kinetics of RB19 were analyzed using the Langmuir-Hinshelwood model. Fig. 7b illustrates a linear relationship of $\ln(C_t/C_0)$ against time, exhibiting high $R^2$ values ranging from 0.9396 to 0.9891. These elevated $R^2$ values suggest conformity to pseudo-first-order kinetics. The rate constants ($k$) for $Ag_2MoO_4$, $Ag_2MoO_4$/CF, $Cu_2O$, $Cu_2O$/$Ag_2MoO_4$, and $Cu_2O$/$Ag_2MoO_4$/CF were determined to be 3.69 x $10^{-3}$, 6.21 x $10^{-3}$, 5.54 x $10^{-3}$, 7.84 x $10^{-3}$, and 28.12 x $10^{-3}$ $min^{-1}$, respectively. Remarkably, $Cu_2O$/$Ag_2MoO_4$/CF exhibited significantly higher catalytic activity than those of the pristine $Ag_2MoO_4$, $Cu_2O$, and their heterojunctions, surpassing these photocatalysts by 3.6–7.6 times. Furthermore, compared to several recently published photocatalytic materials for RB19 degradation listed in Table 1, $Cu_2O$/$Ag_2MoO_4$/CF demonstrates comparable catalytic activity, underscoring its potential practical application.

Fig. 7c illustrates the time-dependent degradation spectra of RB19 under visible light in the presence of $Cu_2O$/$Ag_2MoO_4$/CF. As time progresses, a noticeable reduction in the peak intensity can be observed. This degradation of the RB19 chromophore is responsible for the observed color change and the decrease in absorption spectra. Furthermore, the mineralization of RB19 was assessed by measuring the TOC content, as depicted in Fig. 7d. Remarkably, 87.8 % of TOC was successfully removed, which closely corresponds to the 98.9 % degradation observed over 150 min. The significant mineralization achieved suggests the effective breakdown of RB19 into simpler and less harmful byproducts, highlighting the efficiency of the photocatalytic process.

### 3.3. Effects of operating parameters and reusability

Comprehending the influence of variables like catalyst dosage, RB19 concentration, and pH levels on photocatalytic performance is crucial for enhancing the real-world applicability of $Cu_2O$/$Ag_2MoO_4$/CF. In these experiments, $C_0$ represents the RB19 concentration after 60 min of adsorption. The relationship between catalyst dosage and photocatalytic efficiency was investigated at various catalyst dosages (0.5, 1.0, 2.0, and 3.0 g/L), as presented in Fig. 8a. It is observed that increasing the catalyst dosage from 0.5 to 2.0 g/L leads to an enhancement in photocatalytic efficiency from 57.6 % to 99.5 % within 150 min of illumination. This improvement can be attributed to the increased number of active sites provided by the catalyst's surface as the dosage is increased [47]. However, when the catalyst dosage is further increased to 3.0 g/L, the RB19 degradation efficiency decreases to 73.5 %. This phenomenon may be attributed to an excess of catalyst, which could shield RB19 and light from reaching the catalytic centers [44]. Although the highest degradation efficiency is achieved at a 2.0 g/L catalyst dosage, the efficiency at 1.0 g/L is also comparable (97.1 %). Therefore, 1.0 g/L is considered the most effective catalyst dosage.

**Table 1**
Comparison of catalytic activity of various catalysts toward RB19.

| Catalyst | Type of reaction | Experimental conditions ($C_0$ (mg/L)/ catalyst dosage (g/L)/pH | DE(%)/k ($min^{-1}$) | Ref. |
|---|---|---|---|---|
| $Cu_2O$/$Fe_3O_4$/UiO-66 | Fenton | 100/1.6/7 | 83.9 /0.019 | [42] |
| $CdIn_2S_4$/g-$C_3N_4$ | Photo-Fenton | 20/1/3 | ~100/ 0.073 | [43] |
| $Fe_3O_4$/$MnO_2$ | Photocatalytic | 100/0.3/7 | 62.5/ 0.015 | [44] |
| $Bi_2MoO_6$/$Bi_2WO_6$/MWCNTs | Photocatalytic | 50/0.6/- | 96.9/ 0.013 | [45] |
| α-$Fe_2O_3$/$WO_3$ | Photocatalytic | 30/1/2 | 83/- | [46] |
| $BiPO_4$/g-$C_3N_4$ | Photocatalytic | 20/1/3 | 70.1/ 0.016 | [47] |
| $TiO_2$/$KBrO_3$ | Photocatalytic | 20/1/7 | 98.9/- | [48] |
| $Cu_2O$/$Ag_2MoO_4$/CF | Photocatalytic | 25/1/7 | 98.9/ 0.028 | This study |

The influence of pH on the RB19 degradation by $Cu_2O$/$Ag_2MoO_4$/CF was investigated within the pH range of 3–9 under RB19 concentration of 25 mg/L and a catalyst dosage of 1 g/L. The experimental results, as depicted in Fig. 8b, unequivocally affirm that the initial pH of the RB19 solution significantly impacts the catalytic activity of the material. The RB19 degradation efficiency after 150 min of reaction is found to be 58.8 %, 76.9 %, 97.1 %, and 47.8 % at pH values of 3, 5, 7, and 9, respectively. This trend suggests a distinct pH-dependent behavior. The observed behavior can be attributed to the point of zero charge (PZC) of $Cu_2O$/$Ag_2MoO_4$/CF, which is determined to be 7.6 (as presented in Fig. S1 of supplementary information). Below the $pH_{PZC}$ (pH 3), the catalyst's surface becomes positively charged, enhancing the adsorption of negatively charged RB19 molecules. While this stronger adsorption theoretically favors photocatalytic activity, it can reach a point where excessive adsorption creates a dense RB19 layer on the catalyst's surface. This layer, in turn, limits the penetration of light into the catalyst, reducing the overall efficiency of the photocatalytic process [40]. Conversely, at pH values above the $pH_{PZC}$, the catalyst's surface becomes negatively charged, leading to repulsion between the catalyst and RB19 molecules, resulting in reduced adsorption capacity. However, the relatively higher pH values (pH 7) bring the surface charge closer to neutrality, optimizing the catalyst's performance by maintaining a balance between adsorption and photocatalytic activity.

The influence of the initial concentration of RB19 on $Cu_2O$/$Ag_2MoO_4$/CF photocatalysis is crucial due to the varying pollutant concentrations in industrial wastewater. Experiments were conducted at different initial RB19 concentrations (15, 25, 35, and 45 mg/L) while maintaining a constant pH of 7 and a catalyst dosage of 1 g/L. Fig. 8c reveals a significant trend: as the initial RB19 concentration increases, the photocatalytic efficiency of $Cu_2O$/$Ag_2MoO_4$/CF decreases. This decrease in efficiency can be attributed to several factors. At higher RB19 concentrations, competitive adsorption onto the catalyst's active sites intensifies, reducing interactions between RB19 molecules and active sites and impeding the photocatalytic process [49]. Mass transfer limitations may also occur at elevated RB19 concentrations due to the high concentration of RB19 molecules in the bulk solution, decreasing the contact time between RB19 molecules and the catalyst's surface [50]. Additionally, beyond a certain RB19 concentration threshold, the catalyst's active sites may become saturated with RB19 molecules, leading to diminishing returns in terms of photocatalytic efficiency. Furthermore, at higher RB19 concentrations, the dye molecules can absorb a significant portion of incident light, limiting the amount of light available for photocatalytic reactions [44]. This light absorption phenomenon can result in reduced photocatalytic efficiency as the RB19 concentration increases.

The long-term stability and reusability of the $Cu_2O$/$Ag_2MoO_4$/CF composite are critical considerations for its practical utility. To assess its stability, a series of RB-19 degradation cycles was performed, as illustrated in Fig. 8d. The results demonstrate that even after undergoing five successive degradation cycles, the photocatalytic activity of $Cu_2O$/$Ag_2MoO_4$/CF remains remarkably consistent, with the RB-19 degradation rate consistently exceeding 86.48 %.

To further evaluate the stability, the leaching of metal ions, specifically Cu and Ag, was monitored during the photocatalytic reactions. The concentration of leached metals across multiple reuse cycles is depicted in Fig. 8e. Notably, there is a gradual decrease in the leached metal concentrations with each successive cycle. This phenomenon can be attributed to the fact that surface-bound metals are initially more prone to leaching in the early cycles due to their higher availability for interaction with the reacting species. Noticeably, the total amount of Cu and Ag leached into the solution over five reuse cycles was only 0.182 mg/L and 0.025 mg/L, respectively. Considering the chemical composition of the material's surface, as determined by EDX analysis (Fig. 5),

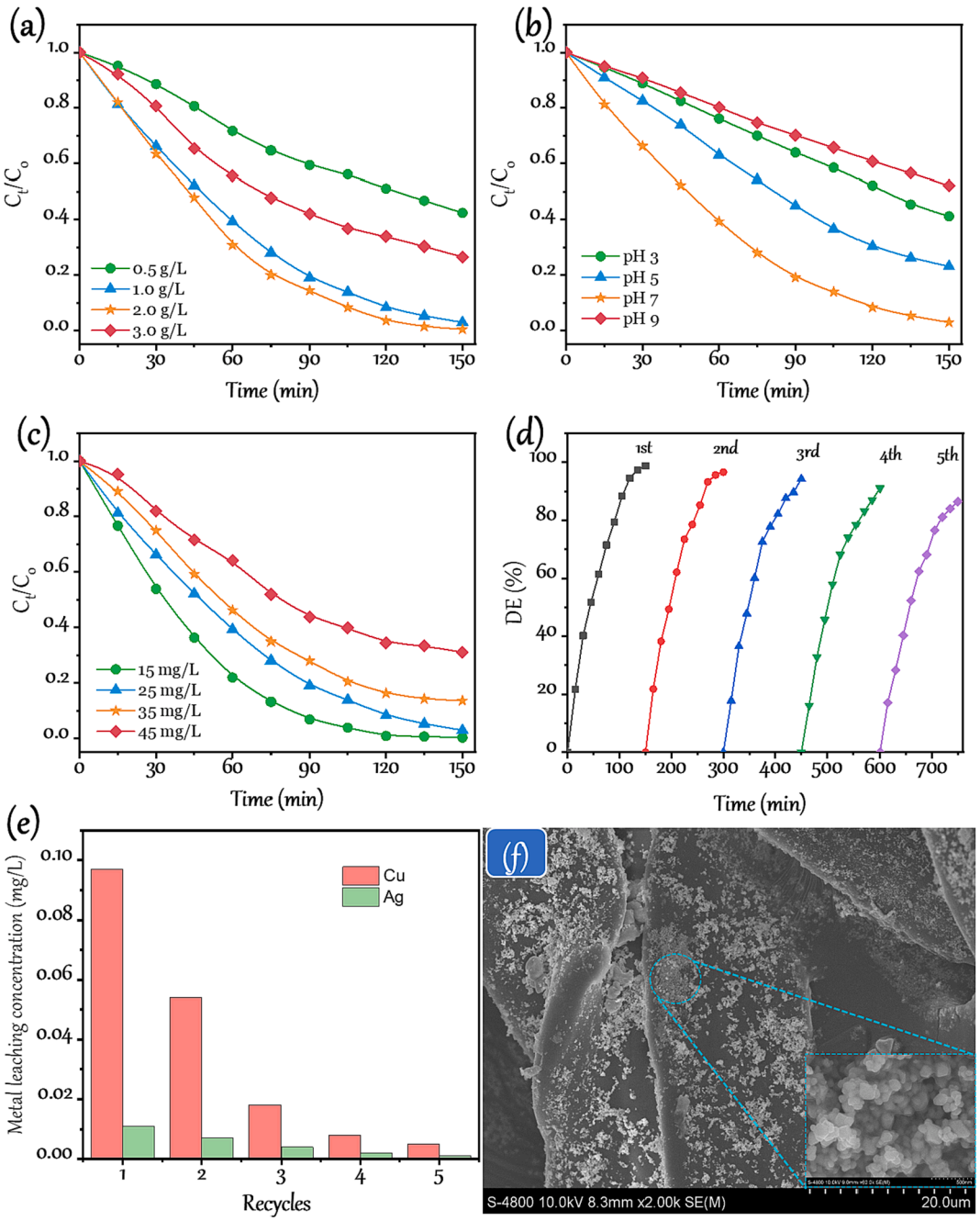



**Fig. 8.** Effect of catalyst dosage (a), solution pH (b), and RB19 concentration (c), DE of RB19 (d) and leaching concentration of Cu and Ag in recycles (e), SEM image of $Cu_2O/Ag_2MoO_4/CF$ after reuse.

the leached Cu and Ag accounted for approximately 0.15 % and 0.08 %, respectively, of the initial Cu and Ag content in the composite. This achievement can be attributed, in part, to the neutral conditions under which the photocatalytic process was conducted. Importantly, the concentrations of leached Cu and Ag ions remained consistently low throughout the reuse cycles and remained well below the permissible limits for Cu (2 mg/L) and silver (0.1 mg/L) in drinking water, as established by the World Health Organization (WHO).

Fig. 8f provides a closer look at the $Cu_2O/Ag_2MoO_4/CF$ photocatalyst's morphology after five cycles of RB-19 photodegradation, as observed through FE-SEM. Notably, no significant changes in the morphology of the $Cu_2O/Ag_2MoO_4/CF$ composite are observed. The nano-sized $Cu_2O$ and $Ag_2MoO_4$ catalyst particles maintain strong adhesion to the CFs substrate. Collectively, these findings affirm that the $Cu_2O/Ag_2MoO_4/CF$ heterojunction exhibits exceptional stability and holds substantial promise for practical applications in various production processes.

### 3.4. Possible photocatalytic degradation mechanism

Before proposing a suitable mechanism for RB19 degradation, the influence of radical scavengers within the photodegradation system was investigated. Specific scavengers (1 mM) such as benzoquinone (*p*-BQ), isopropyl alcohol (IPA), EDTA, and $K_2Cr_2O_7$ were employed to trap superoxide radicals ($^{\bullet}O_2^-$), hydroxyl radicals ($^{\bullet}OH$), holes ($h^+$), and electrons ($e^-$), respectively [51,52]. As depicted in Fig. 9a, the presence of scavengers had a notable impact on the degradation efficiency of $Cu_2O/Ag_2MoO_4/CF$. Specifically, the degradation efficiency dropped

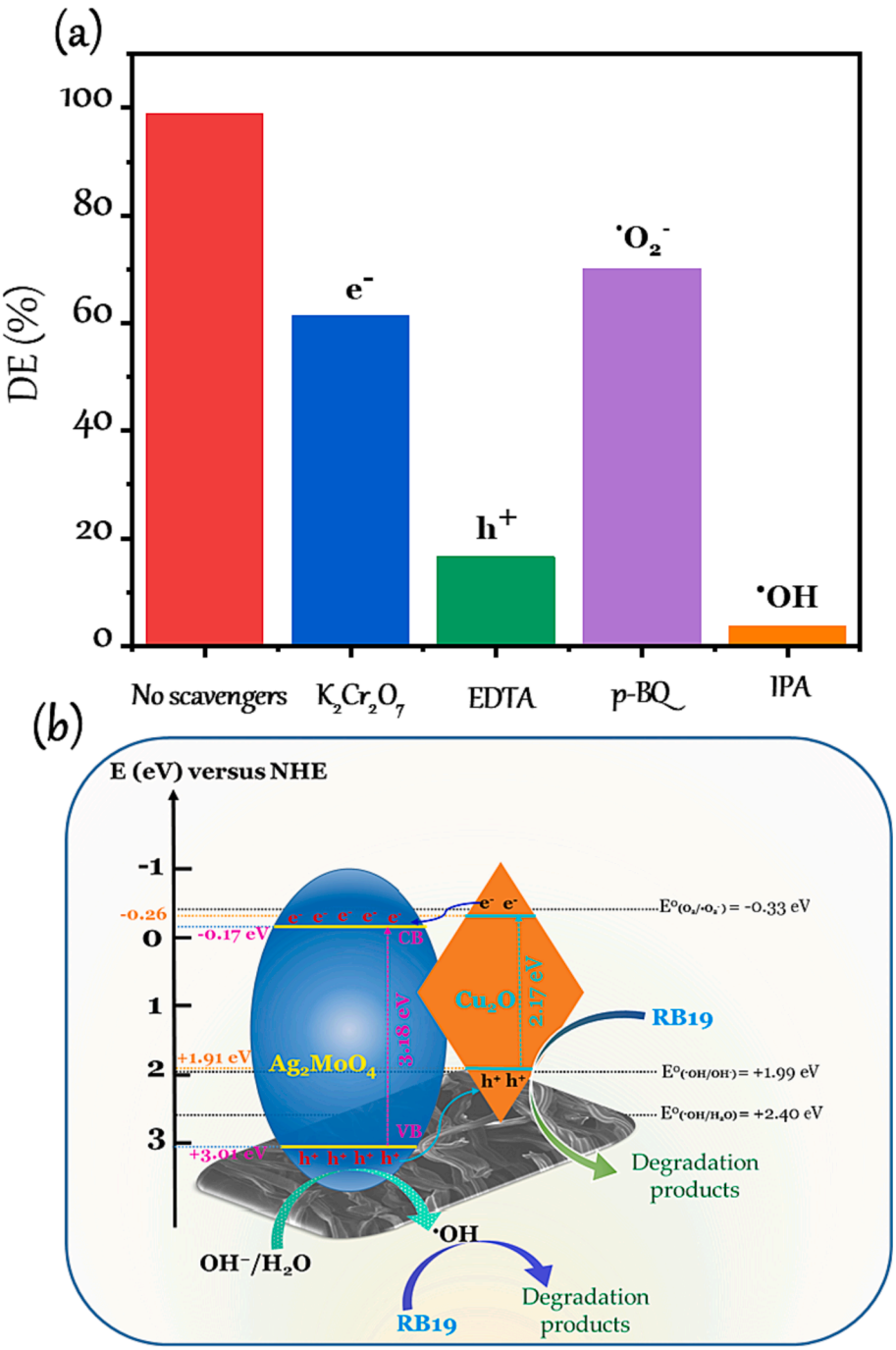



**Fig. 9.** Effect of scavengers (a) and possible mechanisms of the ternary type II $Cu_2O/Ag_2MoO_4$/CF heterojunctions (b) for the RB19 degradation.

from 98.9 % to 70.2 % in the presence of $^{\bullet}O_2^-$ scavenger, to 61.5 % with $e^-$ scavenger, to 16.7 % with $h^+$ scavenger, and to 3.8 % with $^{\bullet}OH$ scavenger. These results strongly suggest that the primary active species involved in the RB19 degradation process were $^{\bullet}OH$ and $h^+$.

To elucidate the charge transfer pathways, we specified the energy band positions of the valence band (VB) and conduction band (CB) for each semiconductor. The calculated energy band positions for $Cu_2O$ were −0.26 eV (CB) and + 1.91 eV (VB), while for $Ag_2MoO_4$, they were −0.17 eV (CB) and + 3.01 eV (VB). Considering that the reduction potential of $O_2/^{\bullet}O_2$ – is approximately − 0.33 eV, which is more negative than the CB potential of $Cu_2O$ and $Ag_2MoO_4$, it can be inferred that the CB electrons of both catalysts cannot effectively reduce $O_2$ to $^{\bullet}O_2^-$ radicals [14]. The VB potential of $Ag_2MoO_4$ (+3.01 eV) is significantly more positive than the standard oxidation potentials of $H_2O/^{\bullet}OH$ (+2.40 eV) and $OH^-/^{\bullet}OH$ (+1.99 eV). This observation implies that the photo-induced holes generated in $Ag_2MoO_4$ VB band can directly oxidize $OH^-$ and $H_2O$, yielding a substantial quantity of $^{\bullet}OH$ radicals [5]. Concurrently, due to the energy band potential difference, a portion of the $h^+$ from $Ag_2MoO_4$ will migrate to the VB band of $Cu_2O$. The observed migration of photo-excited electrons and holes strongly supports the categorization of the $Cu_2O/Ag_2MoO_4$/CF composite as a type-II heterojunction [53]. This migration enables direct oxidation of RB19 and reduces the recombination rate of electron-hole pairs, aligning well with the results of radical quenching experiments. Based on the foregoing results, a possible type-II heterojunction charge transfer mechanism for RB19 degradation by $Cu_2O/Ag_2MoO_4$/CF is illustrated in Eqs. (6)-(10) and Fig. 9b.

$$Cu_2O/Ag_2MoO_4/CF + h\nu \rightarrow Cu_2O/Ag_2MoO_4/CF(e^-_{(CB)} + h^+_{(VB)}) \quad (6)$$

$$h^+(Ag_2MoO_4) + OH^- \rightarrow {}^\bullet OH \quad (7)$$

$$h^+(Ag_2MoO_4) + H_2O \rightarrow H^+ + {}^\bullet OH \quad (8)$$

$${}^\bullet OH + RB19 \rightarrow \text{Degradation products} \quad (9)$$

$$h^+(Cu_2O) + RB19 \rightarrow \text{Degradation products} \quad (10)$$

## 4. Conclusions

The successful synthesis of $Cu_2O/Ag_2MoO_4$ nanoheterojunctions supported on CFs has been rigorously confirmed through a comprehensive series of characterizations. This novel composite has exhibited remarkable photocatalytic prowess in the degradation of RB19 under visible light. Notably, the ternary $Cu_2O/Ag_2MoO_4/CF$ composite has outperformed its constituent parts, $Cu_2O$, pristine $Ag_2MoO_4$, or $Cu_2O/Ag_2MoO_4$, by a substantial margin, achieving degradation rates 3.6–7.6 times higher. This remarkable enhancement can be primarily ascribed to the composite's enhanced visible-light absorbance and its ability to effectively separate and transfer photo-generated electron/hole ($e^-/h^+$) pairs. Following 150 minof visible light irradiation, the 1.0 g/L $Cu_2O/Ag_2MoO_4/CF$ photocatalyst successfully degraded 98.9 % of the 25 mg/L RB19 dye solution at pH 7. Furthermore, the research has underscored the substantial influence of pH and catalyst dosage on the catalytic activity of the ternary composite. The outcomes of free radical capture experiments have unequivocally highlighted the pivotal roles played by hydroxyl radicals ($^\bullet OH$) and holes ($h^+$) in the photocatalytic process, whereas superoxide radicals ($^\bullet O_2^-$) exhibited negligible effects. Furthermore, reusability experiments have convincingly demonstrated the composite's robustness and its potential for repeated utilization. Even after undergoing five consecutive cycles, the composite maintained a photocatalytic efficiency exceeding 86 %. In summary, the $Cu_2O/Ag_2MoO_4/CF$ ternary composite emerges as a highly promising photocatalyst for the degradation of RB19. It boasts superior visible-light photocatalytic performance, durability, and the potential for widespread applications in wastewater treatment and environmental remediation.

## CRediT authorship contribution statement

**Van-Dat Doan:** Writing – original draft, Validation, Methodology, Investigation, Methodology, Software. **Thi Thanh Ngan Nguyen:** Data curation, Writing – original draft. **Hoang Ai Le Pham:** Visualization, Investigation. **Thi Lan Huong Nguyen:** Investigation, Methodology. **Olga Evgenievna Lebedeva:** Software, Validation, Methodology. **Huu Phuc Dang:** Resources, Visualization, Writing – original draft. **Anh-Tien Nguyen:** Data curation, Formal analysis. **Vy Anh Tran:** Resources, Software. **Van Thuan Le:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.molliq.2024.124261.

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